

# CAREERX - Agentic AI Career Navigator

## Problem Statement

### The Challenge

Industry hiring requirements are constantly changing, and students often struggle with career planning, identifying skill gaps, and mapping out a learning path. Information about industry requirements, specific job role skills, and necessary projects is scattered and vague. Generic advice fails to account for a student's existing knowledge, leading to skill mismatches and wasted time on unnecessary certifications. Without personalized, real-time guidance, many fail to align their learning with actual industry demands or make informed career choices.

### The Objective

Build an AI-powered CAREERX that delivers personalized learning insights and guidance for students, freshers, and university placement cells. The solution aims to bridge the gap between academia and industry requirements by dynamically mapping student skills to job market demands.

### Key Capabilities Required

1. **Personalized Learning Roadmaps:** Recommend step-by-step weekly plans based on existing skills, available study hours, and the chosen career path.
2. **Career Data RAG Retrieval:** Fetch and summarize industry role requirements from a curated knowledge base ensuring recommendations are up-to-date.
3. **Skill Gap Analysis:** Compare student skills to industry benchmarks and provide priority-based gap analysis, highlighting what is urgently needed.
4. **Project Recommendation Focus:** Suggest specific projects and certifications to close skill gaps (e.g., building a full-stack project or obtaining an AWS certification).

### Multi-Agent Approach

To achieve this, the system leverages a multi-agent architecture including:

## **CAREERX - Agentic AI Career Navigator**

- Career Research Agent: Retrieves and summarizes career information using RAG.
- Skill Gap Agent: Identifies missing skills by comparing user profiles against requirements.
- Learning Path Agent: Generates personalized, step-by-step roadmaps.
- Project & Cert Agent: Provides tailored project recommendations that directly close identified gaps.